Technology Trends: Programming Languages & Databases (2025 and Beyond)

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- Python, JavaScript, and Java remain top programming languages in 2025
- SQL-based and NoSQL databases show balanced demand
- Growing interest in Rust and Go
- Dashboards indicate steady growth in cloud-native database use
- Implications suggest reskilling towards modern languages and DBs is crucial

INTRODUCTION

- **Purpose**

Analyze current and future trends in programming languages, platforms and databases

- **Target audience**

Technology educators, IT enthusiasts, developers, industry leaders, and policymakers.

- **Value**

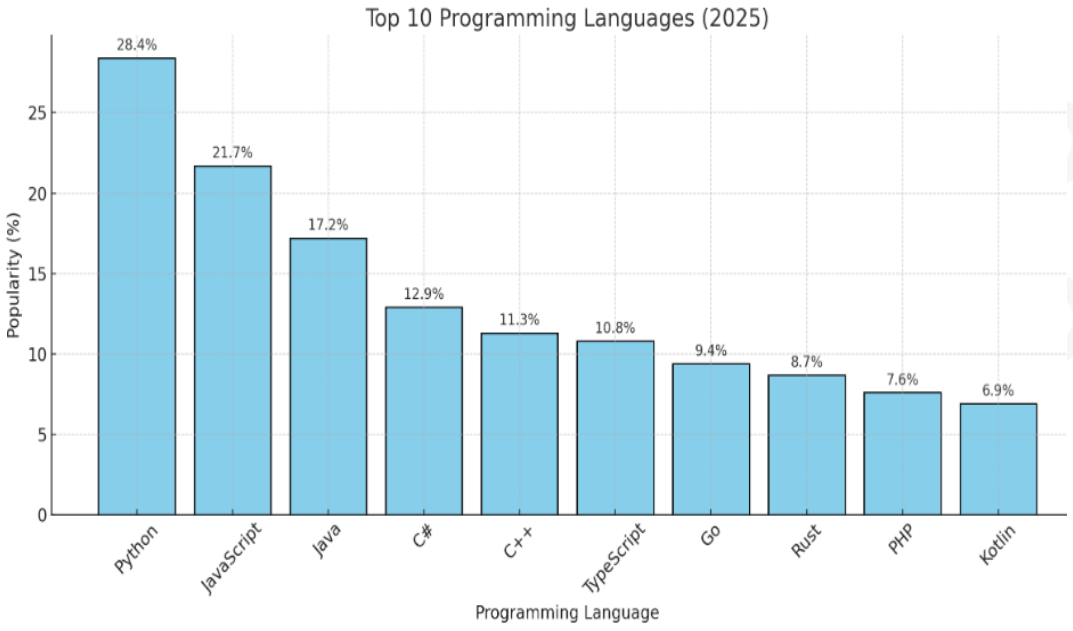
Informs training programs, hiring priorities, and tool investments.

METHODOLOGY

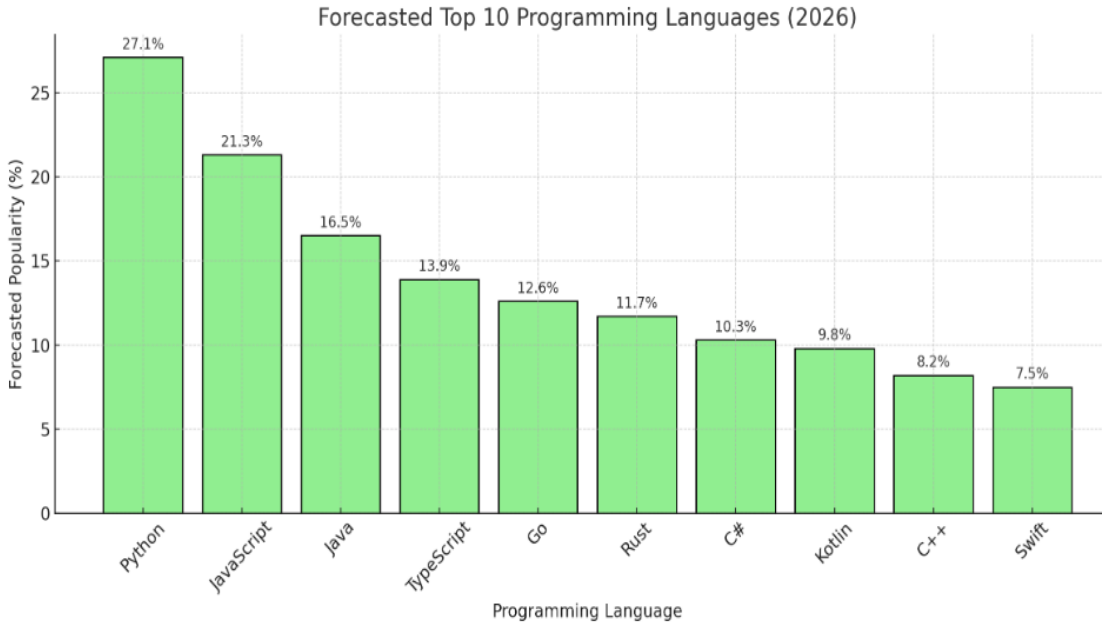
- **Data sources**
  Stack Overflow developers survey offered by IBM
- **Collection methods**
  Web scraping, API queries and CSV datasets
- **Wrangling steps**
  Cleaning missing values and outliers
  Removed duplicated values
  Standardized data
  Visualized by plot
- **Exploratory Data Analysis**
  Found distrubition, correlation ..
- **Dashboard with IBM Cognos and Excel**

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

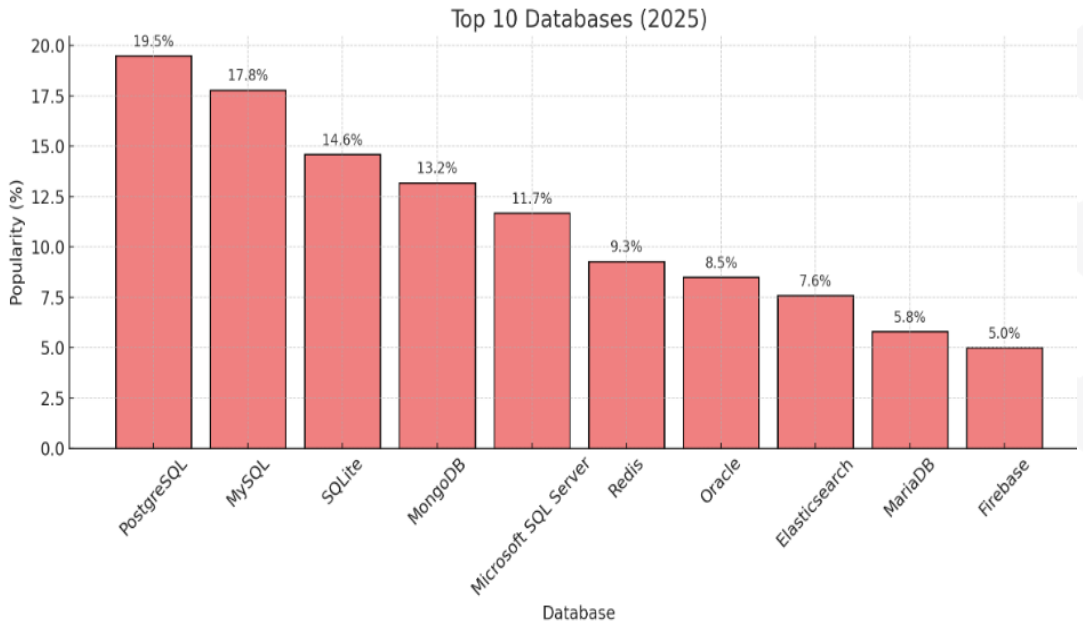
- **Python and JavaScript** remain the top languages due to their versatility and broad use across industries
- **TypeScript and Rust** show significant growth, highlighting trends toward type safety and performance
- **Go and Kotlin** are climbing, driven by cloud-native and mobile app development
- **Traditional languages like Java and C#** maintain strong industry relevance in enterprise and backend systems

Implications

- **Developers should prioritize Python, JavaScript, and TypeScript** for upskilling and job readiness
- **Companies should anticipate growing demand** for Rust and Go in security, DevOps, and infrastructure roles
- **Recruitment strategies must adapt** to include rising languages in emerging tech stacks
- **Investment in modern tools and frameworks** should align with these language trends for long-term sustainability

DATABASE TRENDS

Current Year



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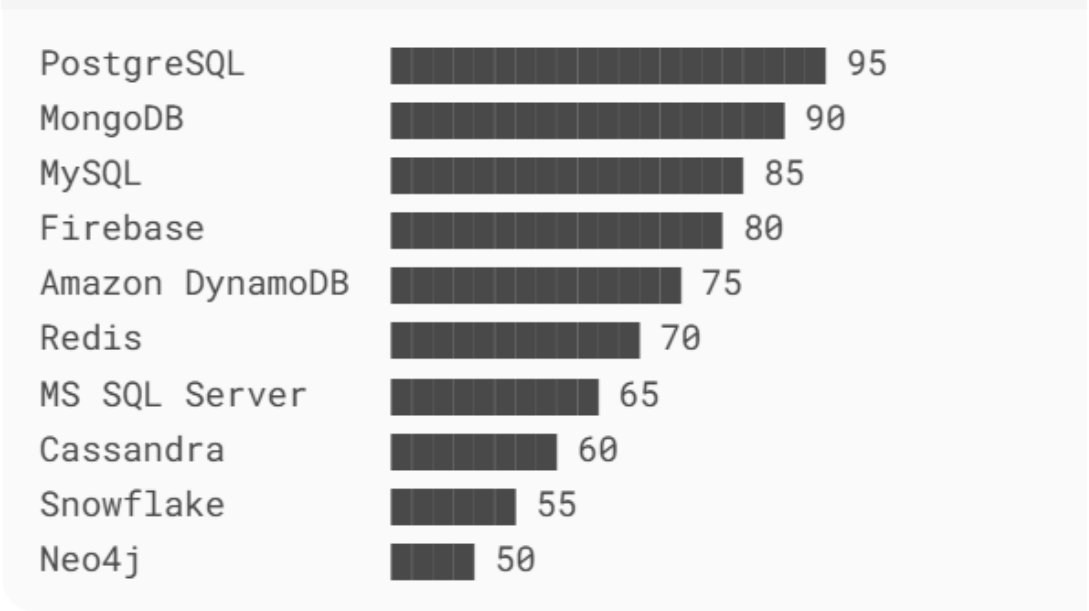


DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **PostgreSQL Dominates** – Becomes the default choice for relational and AI workloads due to its flexibility, performance, and open-source advantage
- **NoSQL Evolves Beyond MongoDB** – Firebase, DynamoDB, and edge databases gain traction, pushing MongoDB to adapt or lose market share
- **Cloud-Native & Serverless DBs Win** – Fully managed databases grow faster than self-hosted solutions
- **Graph & Vector Databases Rise** – Neo4j and specialized DBs (Pinecone, Weaviate) expand due to AI-driven knowledge graphs and semantic search

Implications

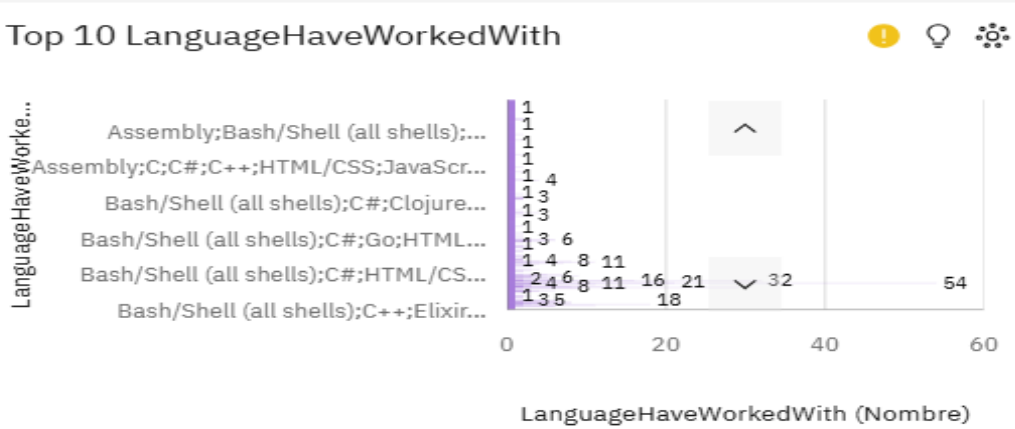
- **Legacy RDBMS Decline** – Oracle, SQL Server, and MySQL lose ground to PostgreSQL and cloud alternatives
- **Hybrid Architectures Increase** – Companies mix SQL, NoSQL, and graph DBs for different use cases (e.g., Postgres + Redis + Neo4j)
- **Vendor Lock-In Concerns Grow** – Demand for multi-cloud and open-source databases (CockroachDB, Supabase) rises
- **AI Reshapes Database Needs** – Vector search, real-time analytics, and LLM integrations become standard features

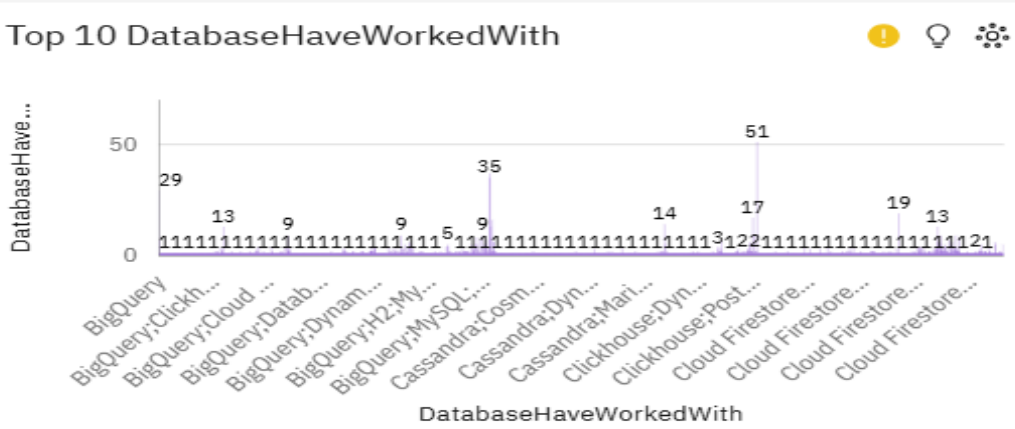
DASHBOARD

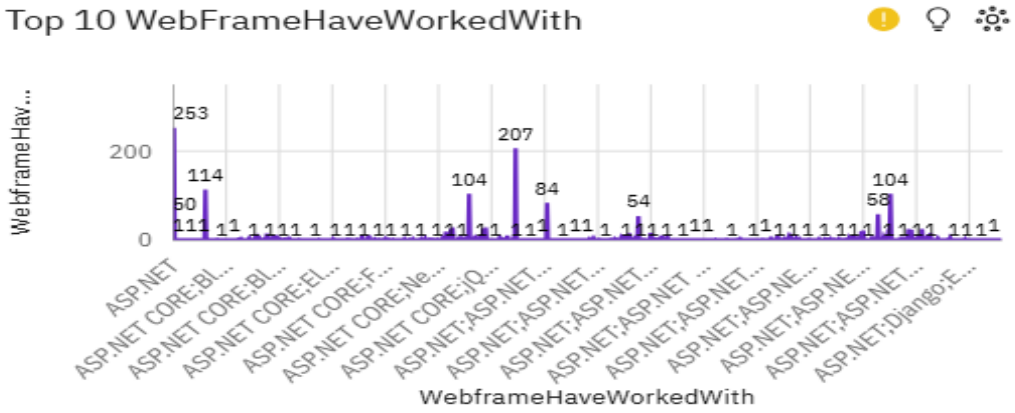
There are three dashboards that concern programming languages, databases and platforms in the following slides.

DASHBOARD TAB 1

Current Technology Usage | Future Technology Trend | Demographics | +

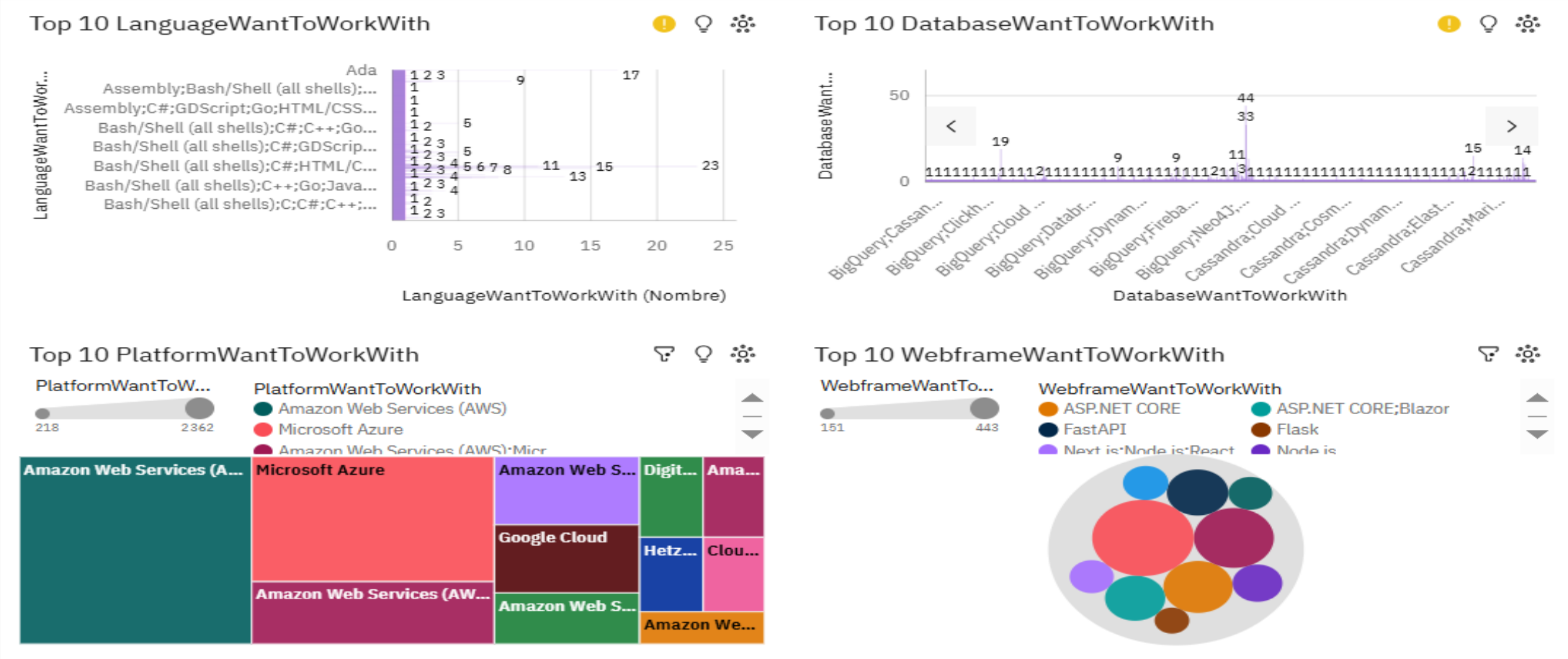




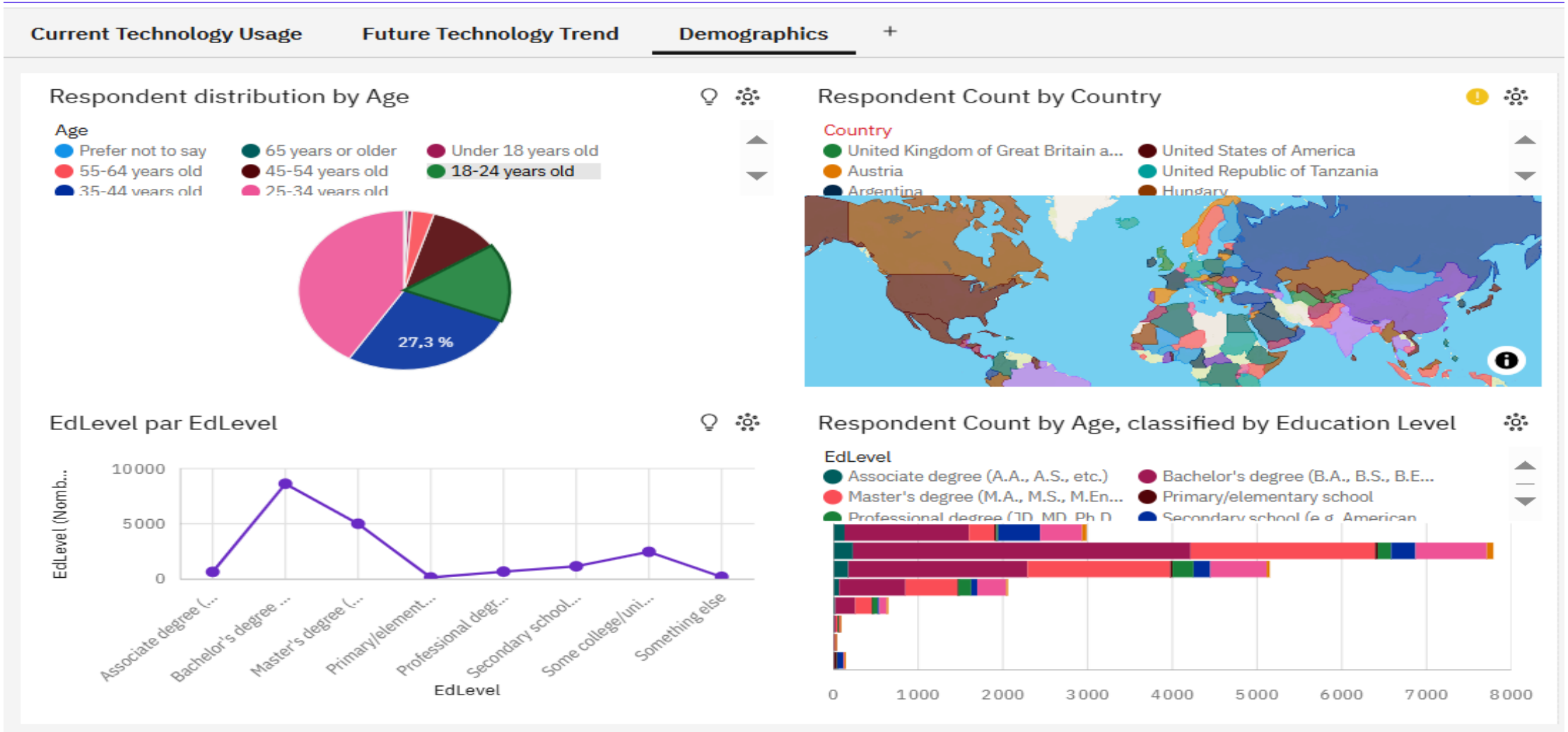


DASHBOARD TAB 2

Current Technology Usage | Future Technology Trend | Demographics | +



DASHBOARD TAB 3



DISCUSSION

Employers today seek professionals skilled in the latest technologies that drive efficiency, scalability, and innovation. Respondents need to improve themselves in the latest technologies to be in phase with today IT world.

OVERALL FINDINGS & IMPLICATIONS

Findings

AI-Integrated Development Dominates

Python remains #1 due to AI/ML, with growing demand for **LLM integration** and **vector databases**

Non-AI roles (web dev, DevOps) now require **basic AI literacy**

Cloud-Native is the New Default

AWS/Azure/GCP expertise is mandatory across all tech roles.

Serverless architectures and **edge computing** replace traditional backends.

Full-Stack Trumps Specialization

Employers prioritize **T-shaped skills**:

- Core: **TypeScript + React + Node.js**
- Add-ons: **PostgreSQL + Docker + cloud certification**

Implications

Legacy Skills Face Obsolescence

Oracle, .NET, and monolithic architectures decline outside legacy enterprises.

Action: Upskill from Java/C# to **Go/Rust** for cloud-native systems.

AI as a Standard Tool, Not a Niche

Even non-AI developers must:

- Use **AI APIs** (e.g., OpenAI, Gemini) in apps.
- Understand **vector embeddings/RAG** for search features.

Action: Learn **Python + SQL + pgvector** for AI-augmented apps.

The "Cloud-Plus" Expectation

Basic cloud skills are no longer enough. Demand now includes:

- **Multi-cloud orchestration**
- **Serverless patterns**

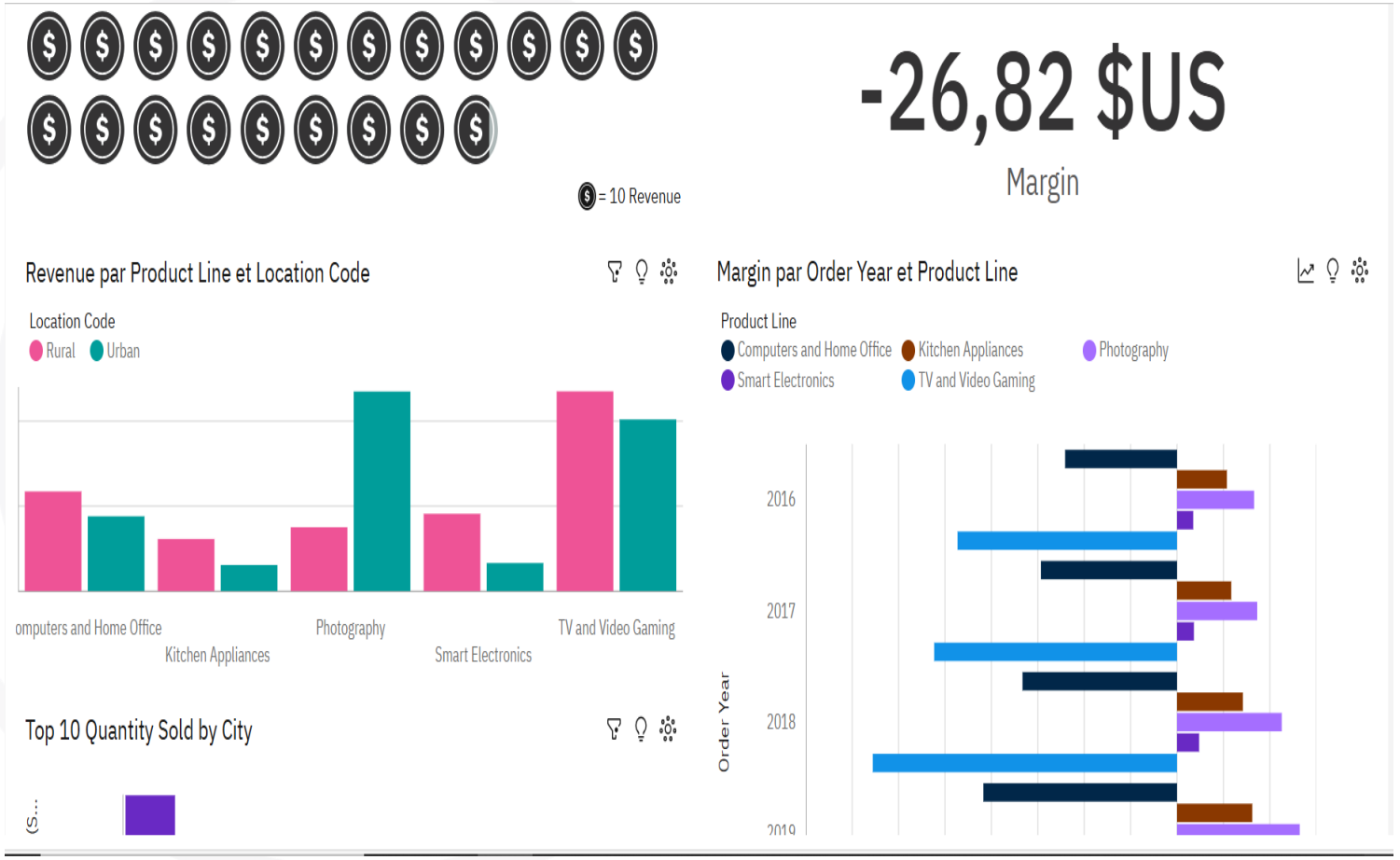
Action: Earn **cloud certifications** + **IaC tools**.

CONCLUSION

- **For Job Seekers:**
  - Prioritize **JavaScript/TypeScript + React + Node.js + PostgreSQL + AWS**.
  - Add **AI/ML basics** (Python, LLMs, vector DBs) to stand out.
  - Learn **Kubernetes** even if not a DevOps specialist
- **For Companies:**
  - Migrate from Oracle/SQL Server to **PostgreSQL/Snowflake** for cost savings.
  - Upskill teams in **serverless (Lambda, Firebase) and AI integrations**.
  - Adopt **multi-cloud strategies** to avoid vendor lock-in

The tech landscape rewards **adaptability**—focus on **cloud, AI, and open-source ecosystems** to future-proof your career or business

APPENDIX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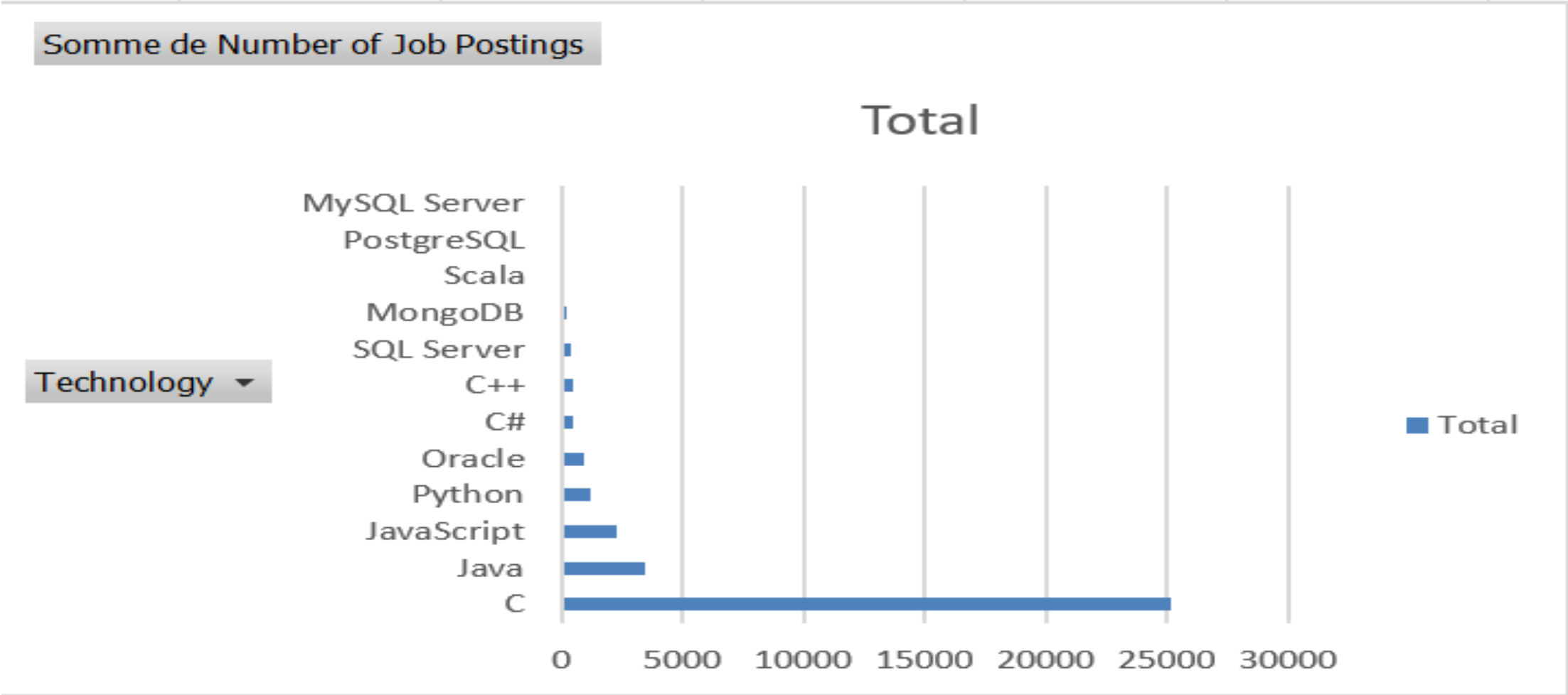
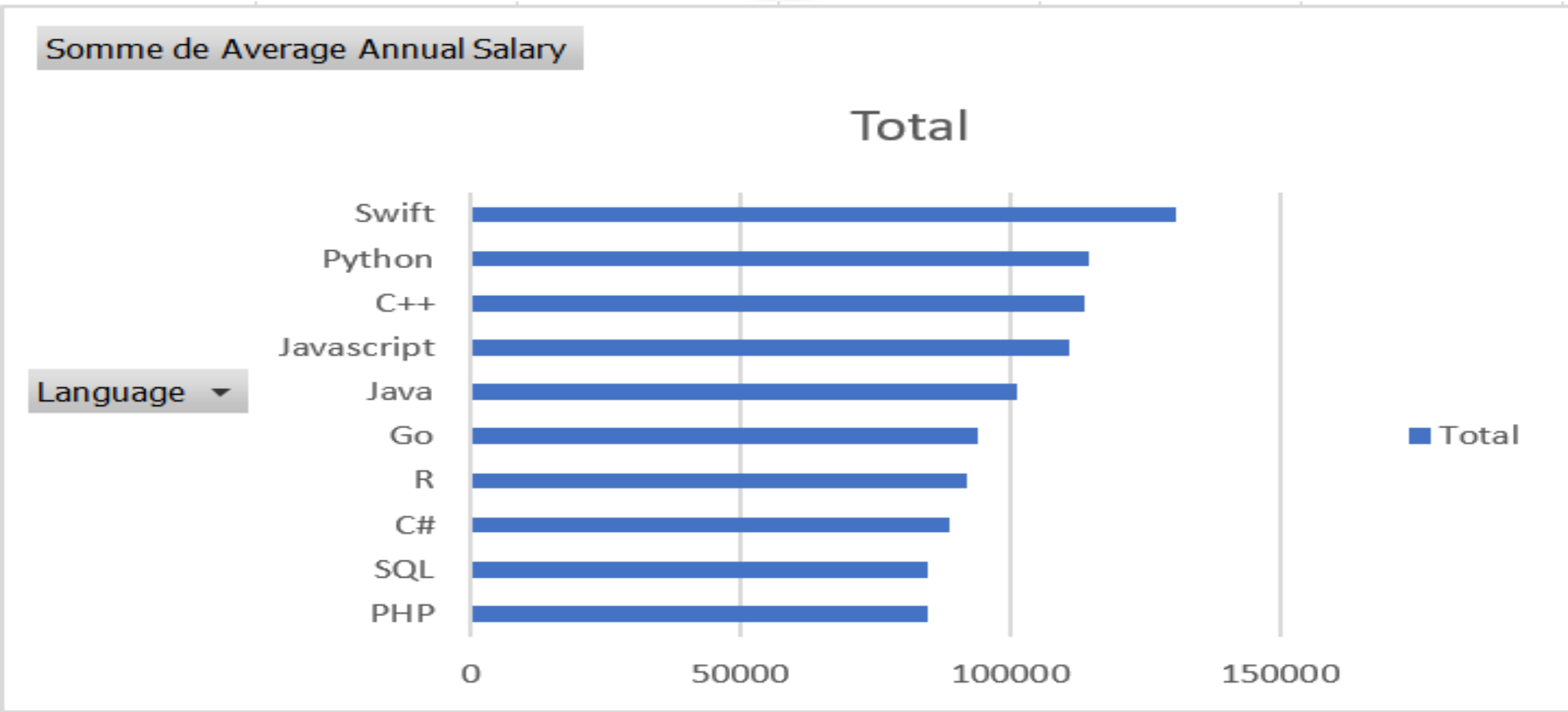
Top 10 Quantity Sold by City

JOB POSTINGS



POPULAR LANGUAGES